وثيقة المراجعة والتلخيص الأكاديمي

Status: READY FOR STUDENT REVIEW AFTER TARGETED CROSS-AUDIT (2026-10-19)
Course: Advanced Software Engineering (CS603) — Asst. Prof. Dr. Ali Fahim Ni'ma
Primary Authority: K.K. Aggarwal & Yogesh Singh, *Software Engineering* (3rd ed.), printed pp. 203–249 (spreads 107–130); Companion Slides 1–99.
Published Deliverables: Six study-pack PDFs plus the canonical index PDF. Final content and diagram corrections are exported and checked; audit scope and limits are recorded in Section 5.

---

1. Chapter Deliverables & Master Verification Matrix

Six study packs and the canonical index are print-ready; targeted source/diagram checks and final gates are recorded below:

Deliverable Artifact	Source Markdown	Compiled PDF	Pages	Items Covered	Delivery Gate Status
T1 — Multiple-Choice (MCQ)	T1_MCQ.md	T1_MCQ.pdf	**11 pp**	25 MCQs (12 publisher slide MCQs 5.1–5.12 + 13 synthesis MCQs)	**Closed**
T2 — Descriptive & Explain	T2_Descriptive.md	T2_Descriptive.pdf	**16 pp**	17 items (12 textbook exercises + 5 seminar gap items)	**Closed**
T3 — Compare & Analyze	T3_Compare_Analyze.md	T3_Compare_Analyze.pdf	**10 pp**	13 items (11 textbook exercises + 2 seminar gap items)	**Closed**
T4 — Visual Models & Diagrams	T4_Diagrams.md	T4_Diagrams.pdf	**8 pp**	5 selected figures (6 semantic-contract entries; 2× PNG redraws)	**Closed**
T5 — Applied Scenarios & Tools	T5_Apply_Scenarios.md	T5_Apply_Scenarios.pdf	**12 pp**	7 items (direct textbook Ex 5.12 + 6 applied scenarios & diagnostics)	**Closed**
V5 — Vocabulary Coverage	V5_Vocabulary_Coverage.md	V5_Vocabulary_Coverage.pdf	**13 pp**	24 terms from the student-provided professor vocabulary list; provenance is recorded in V5_Doctor_Vocabulary.md	**Closed**
TOTAL CHAPTER OUTPUT	**6 Markdown Packs**	**6 pack PDFs + index PDF**	**70 pack pages + 6 index pages = 76 pages**	**Final gates passed; bounded review**	**Ready for student review**

> 📝 **Academic Note**
>
> **Zero Calculation Pack Invariant:** Unlike Chapter 4, Chapter 5 contains no computational formulas or numerical estimation models. Structural metrics (such as Fan-In and Fan-Out) are structural heuristics evaluated qualitatively in analytical and applied scenarios (`T3` and `T5`). In accordance with `Ch05_BUILD_PLAN.md`, no artificial calculation pack (`T7`) was fabricated.

---

2. Complete Course Question Inventory & Dispositions Ledger

A. Official Slide MCQs (12 Items)

All 12 publisher-slide MCQ stems and answer keys were cross-checked against slides 94–96 and the cited course pages, then accounted for in `T1_MCQ.md`.

MCQ	Actual Question Stem	Pack Location	Verified Key	Source Disposition
`5.1`	The most desirable form of coupling is:	`T1_MCQ.md#Q5.1`	**(b) Data Coupling**	Verified: Book p. 208; Slide 16, 94.
`5.2`	The worst type of coupling is:	`T1_MCQ.md#Q5.2`	**(a) Content coupling**	Verified: Book p. 210; Slide 24, 94.
`5.3`	The most desirable form of cohesion is:	`T1_MCQ.md#Q5.3`	**(c) Functional cohesion**	Verified: Book p. 212; Slide 30, 94.
`5.4`	The worst type of cohesion is:	`T1_MCQ.md#Q5.4`	**(b) Coincidental cohesion**	Verified: Book p. 211; Slide 26, 94.
`5.5`	Which one is not a strategy for design?	`T1_MCQ.md#Q5.5`	**(c) Embedded design**	Verified: Book pp. 213–215; Slide 32, 95.
`5.6`	If all task must be executed in the same time-frame, what type of cohesion is being exhibited?	`T1_MCQ.md#Q5.6`	**(c) Cohesion with respect to time**	Verified: Book p. 211; Slide 28, 95.
`5.7`	In functional cohesion, each software module performs:	`T1_MCQ.md#Q5.7`	**(a) Single functional task in same procedure**	Verified: Book p. 212; Slide 30, 95.
`5.8`	Two modules are said to be ... if they refer to the same global data area:	`T1_MCQ.md#Q5.8`	**(d) Common coupled**	Verified: Book p. 209; Slide 22, 95.
`5.9`	What elements of processing are grouped together in a module, in procedural cohesion?	`T1_MCQ.md#Q5.9`	**(c) Procedural cohesion**	Verified: Book p. 211; Slide 28, 96.
`5.10`	The relationship of data elements in a module is called:	`T1_MCQ.md#Q5.10`	**(b) Cohesion**	Verified: Book p. 210; Slide 25, 96.
`5.11`	A system that does not interact with external environment is:	`T1_MCQ.md#Q5.11`	**(a) Closed system**	Verified: Book p. 204; Slide 2, 96.
`5.12`	The extent to which different modules are dependent upon each other is called:	`T1_MCQ.md#Q5.12`	**(a) Coupling**	Verified: Book p. 207; Slide 15, 96.

B. Textbook Exercises (24 Items)

The publisher exercises are mapped across the pack suite; answer wording was corrected in a targeted cross-audit against the cited material. The review was not a blind audit of every sentence:

Exercise	Core Topic & Task	Pack Destination	Primary Question ID
5.1	Define design; contrast conceptual vs technical design	T2_Descriptive, T3_Compare_Analyze, T4_Diagrams	Q5.D01, Q5.D02, Q5.C01, Q5.V01
5.2	Explain design objectives; informal to detailed design	T2_Descriptive	Q5.D03
5.3	Distinguish between design and coding; anti-patterns	T2_Descriptive	Q5.D04
5.4	Define modularity; Bertrand Meyer's five criteria	T2_Descriptive, T5_Apply_Scenarios	Q5.D05, Q5.S03
5.5	Define coupling; explain six coupling levels	T2_Descriptive, T5_Apply_Scenarios	Q5.D06, Q5.S04
5.6	Define cohesion; explain seven cohesion levels	T2_Descriptive, T5_Apply_Scenarios	Q5.D07, Q5.S04
5.7	Analyze cohesion-coupling duality and quality trade-offs	T3_Compare_Analyze	Q5.C04
5.8	Analyze why logical cohesion induces control coupling	T3_Compare_Analyze	Q5.C05
5.9	Analyze operational consequences of high coupling	T3_Compare_Analyze, T5_Apply_Scenarios	Q5.C06, Q5.S04
5.10	Analyze architectural consequences of low cohesion	T3_Compare_Analyze, T5_Apply_Scenarios	Q5.C07, Q5.S04
5.11	Compare Top-Down, Bottom-Up, and Hybrid strategies	T3_Compare_Analyze	Q5.C02
5.12	Reuse strategy selection (Bottom-Up direct answer + Hybrid scenario)	T5_Apply_Scenarios	Q5.S01, Q5.S02
5.13	Compare Flowcharts vs Structure Charts	T3_Compare_Analyze	Q5.C03
5.14	Explain why different design notations are required	T2_Descriptive	Q5.D08
5.15	List function-oriented design techniques & procedure layers	T2_Descriptive	Q5.D09
5.16	Define OOD concepts (object, message, abstraction, class, etc.)	T2_Descriptive	Q5.D10
5.17	Compare Abstract Data Types (ADTs) and Classes	T3_Compare_Analyze	Q5.C09
5.18	Evaluate inheritance without polymorphism	T3_Compare_Analyze	Q5.C10
5.19	Explain improvements of OOD over FOD	T2_Descriptive	Q5.D11
5.20	Guidelines for reusable, high-quality classes (God Class fix)	T2_Descriptive, T5_Apply_Scenarios	Q5.D12, Q5.S05
5.21	Outline simplified OOAD step-by-step process	T2_Descriptive	Q5.D13
5.22	Comprehensive comparison: FOD vs OOD	T3_Compare_Analyze	Q5.C11
5.23	Documents produced upon design completion; IEEE 1016 SDD	T2_Descriptive, T5_Apply_Scenarios	Q5.D14, Q5.S07
5.24	Critical evaluation: Complete decoupling ($Coupling = 0$)	T3_Compare_Analyze	Q5.C08

C. Professor-Seminar Gap Coverage

The specific topics highlighted in the instructor seminar review are fully resolved with primary page anchors:

Seminar Topic	Source & Anchor	Pack Location	Question ID
Abstraction Forms (Procedural, Data, Control)	Pressman 5e §13.4.1 (printed pp. 342–343)	`T2_Descriptive`	`Q5.D15`
Stepwise Refinement (Wirth)	Pressman 5e §13.4.2 (printed pp. 343–344)	`T2_Descriptive`	`Q5.D16`
Information Hiding (Parnas)	Pressman 5e §13.4.4 (printed p. 352)	`T2_Descriptive`	`Q5.D16`
Meyer's Five Modularity Criteria	Pressman 5e §13.4.3 (printed p. 345)	`T2_Descriptive`, `T5_Apply_Scenarios`	`Q5.D05`, `Q5.S03`
Fan-In / Fan-Out Metrics & Depth/Width	Pressman 5e §13.4.5; Mall 4e §5.4	`T3_Compare_Analyze`, `T5_Apply_Scenarios`	`Q5.C12`, `Q5.S06`
Formal Design Review Protocol	Rajib Mall 4e §6.7 (printed pp. 350–351)	`T2_Descriptive`, `T5_Apply_Scenarios`	`Q5.D17`, `Q5.S07`
DFD Transform vs Transaction Analysis	Rajib Mall 4e §6.4.1 (printed pp. 344–349)	`T3_Compare_Analyze`, `T4_Diagrams`	`Q5.C13`, `Q5.V05`

3. Supporting-Source Gap-Check Ledger (24 Terms x 5 Candidate Sources)

In accordance with Section 2.5 of `AGENTS.md`, the 24 terms attributed by the student to the professor were mapped against the course source and candidate supporting books. This ledger records intended/useful coverage; it is not a claim that every cell received an exhaustive page-level audit:

#	Vocabulary Leaf Term	Primary: Aggarwal & Singh	Cand 1: Pressman 5e	Cand 2: Mall 4e	Cand 3: Sommerville	Cand 4: B.B. Agarwal
1	What is Design?	**Sufficient (pp. 203–204)**	Confirmed	Confirmed	Confirmed	Redundant
2	Conceptual vs Technical Design	**Sufficient (pp. 204–205)**	Confirmed	Not used	Confirmed	Redundant
3	Objectives of Design	**Sufficient (pp. 203, 206)**	Confirmed	Confirmed	Confirmed	Redundant
4	Why Design is Important?	**Sufficient (pp. 203–204)**	Confirmed	Confirmed	Confirmed	Redundant
5	Modularity	**Gap filled by Pressman**	**Primary Gap Source (§13.4.3)**	Confirmed	Confirmed	Redundant
6	Module Coupling	**Sufficient (pp. 207–210)**	Confirmed	Confirmed	Confirmed	Redundant
7	Module Cohesion	**Sufficient (pp. 210–213)**	Confirmed	Confirmed	Confirmed	Redundant

#	Vocabulary Leaf Term	Primary: Aggarwal & Singh	Cand 1: Pressman 5e	Cand 2: Mall 4e	Cand 3: Sommerville	Cand 4: B.B. Agarwal
8	Cohesion-Coupling Relationship	**Sufficient (pp. 212–213)**	Confirmed	Confirmed	Confirmed	Redundant
9	Strategy of Design	**Sufficient (pp. 213–215)**	Confirmed	Confirmed	Confirmed	Redundant
10	Bottom-Up Design	**Sufficient (pp. 213–214)**	Confirmed	Confirmed	Confirmed	Redundant
11	Top-Down Design	**Sufficient (pp. 214–215)**	Confirmed	Confirmed	Confirmed	Redundant
12	Hybrid Design	**Sufficient (p. 215)**	Confirmed	Confirmed	Confirmed	Redundant
13	Function Oriented Design	**Sufficient (pp. 216–220)**	Confirmed	Confirmed	Confirmed	Redundant
14	Design Notations	**Sufficient (pp. 216–221)**	Confirmed	Confirmed	Confirmed	Redundant
15	Functional Procedure Layers	**Sufficient (pp. 220–221)**	Confirmed	Confirmed	Confirmed	Redundant
16	IEEE SDD Recommended Practice	**Sufficient (pp. 221–224)**	Not used	Not used	Confirmed	Redundant
17	SDD Scope	**Sufficient (p. 221)**	Not used	Not used	Confirmed	Redundant
18	SDD Definitions	**Sufficient (pp. 221–222)**	Not used	Not used	Confirmed	Redundant
19	Purpose of an SDD	**Sufficient (p. 222)**	Not used	Not used	Confirmed	Redundant
20	SDD Information Content	**Sufficient (pp. 222–223)**	Not used	Not used	Confirmed	Redundant
21	SDD Organisation	**Sufficient (pp. 223–225)**	Not used	Not used	Confirmed	Redundant
22	Object Oriented Design	**Sufficient (pp. 225–226)**	Confirmed	Confirmed	Confirmed	Redundant
23	Basic Concepts	**Sufficient (pp. 226–230)**	Confirmed	Confirmed	Confirmed	Redundant
24	Steps to Analyze & Design OOS	**Sufficient (pp. 231–233)**	Confirmed	Confirmed	Confirmed	Redundant

---

4. Diagram Registry & Verification Status

Five selected figures are embedded in T4. Six contract entries cover those figures, with RMS represented by separate DFD and structure-chart contracts. This is a selected visual set, not exhaustive coverage of every figure in the chapter:

Figure ID	Formal Title & Notation	Source Anchor	Editable HTML Source	Rendered 2× Asset	Semantic Gate
FIG-02	Two-Part Design Process (What vs How)	Slide 5, Fig. 2	CH05-FIG02_Two-Part-Design-Process.html	CH05-FIG02_Two-Part-Design-Process.png	**Passed**
FIG-04	Modularity and Software Cost Curve	Slide 14, Fig. 4	CH05-FIG04_Modularity-and-Software-Cost.html	CH05-FIG04_Modularity-and-Software-Cost.png	**Passed**
FIG-19	Transaction-Centered Structure Chart	Slides 42, 44	CH05-FIG19_Transaction-Centered-Structure-Chart.html	CH05-FIG19_Transaction-Centered-Structure-Chart.png	**Passed**
FIG-23	UML Class Generalization (Shape)	Slide 68, Fig. 23	CH05-FIG23_Shape-Inheritance.html	CH05-FIG23_Shape-Inheritance.png	**Passed**
GAP-RMS	RMS DFD Transform Analysis to Structure Chart	Mall 4e §6.4.1	CH05-GAP_RMS_DFD-to-Structure-Chart.html	CH05-GAP_RMS_DFD-to-Structure-Chart.png	**Passed**

---

5. Audit Status and Quality Gates

The chapter is ready for student review after a targeted, multi-agent cross-audit and remediation. Three read-only reviews covered source claims, diagrams/vocabulary, and print delivery; Codex then checked source pages, corrected the substantiated issues, re-exported the packs, and visually inspected changed pages. This was not a blind claim-by-claim verification of every sentence or every source figure.

- All six study-pack Markdown files and this index pass `note_linter.py --strict`.
- All seven PDFs were freshly exported; the total is **76 pages**, with zero text-empty pages.
- The changed pages and corrected FIG-02, FIG-04, and FIG-19 redraws were visually inspected. Other selected figures retain prior review; the atlas lists chapter figures not reproduced in T4.
- The 24-term list is attributed to the professor by the student; the original group post is not in the vault.
- No claim is made that question counts equal exam weighting or that every sentence received exhaustive independent verification.

Retrieval set

1. **What do the four Functional Procedure Layers represent?** Four increasing levels of detail for specifying a procedure, from purpose and relationships to a procedure body.
2. **Which modules are conditional alternatives in FIG-19?** Module 1, Module 2, and Module 3; Input and Output are direct calls from MAIN.
3. **What are the four SDD design views listed in the course material?** Decomposition, Dependency, Interface, and Detail Description.

Current status: ready for student review; close the study loop with your own reading and corrections.